MODULAR KOSZUL DUALITY: SECTIONS 5–8

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1. The collision residue and the seven faces

The bar differential extracts OPE data at collision divisors. When two points collide, the residue of the bar-intrinsic MC element $\Theta_{\mathcal{A}}$ at the codimension-one stratum $D_{\{1,2\}} \subset \overline{\mathrm{FM}}_2(X)$ defines the *collision residue*:

$$r_{\mathcal{A}}(z) \;:=\; \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}}) \;\in\; \mathrm{End}_{\mathcal{A}}(2)[\![z^{-1}]\!]\,. \tag{1.1}$$

The collision residue is the degree-2, genus-0 projection of $\Theta_{\mathcal{A}}$. Its pole orders are one less than those of the OPE: the logarithmic kernel $d\log(z-w)$ absorbs one power of $(z-w)$. For a chiral algebra with OPE poles of maximal order p, the collision residue has poles of order $p-1$.

1.1. Pole-order dichotomy.

The maximal pole order of the collision residue separates the standard landscape into two regimes:

Pole order	Algebra	Collision	Class
1	Heisenberg $\mathcal{H}_k$	k/z	**G**
1	Kac–Moody $\widehat{\mathfrak{g}}_k$	$\Omega/((k+h^\vee)z)$	**L**
3	Virasoro Vir_c	$(c/2)/z^3 + 2T/z$	**M**
$2N-1$	$\mathcal{W}_N$	poles through $z^{-(2N-1)}$	**M**

This table encodes a structural dichotomy: algebras whose collision residue has at most a simple pole (classes **G** and **L**) have finite shadow depth; only class **G** is Swiss-cheese-formal on the derived-center side. Algebras with higher-order poles (class **M**) have infinite shadow towers and genuinely non-formal A_∞-structure.

The dichotomy is not a property of individual OPE coefficients but of the collision residue as a whole: the $d\log$ absorption converts the OPE's double pole (carrying the metric) and simple pole (carrying the Lie bracket) into a single simple pole (carrying the Casimir tensor) for Kac–Moody, while the Virasoro OPE's quartic pole becomes a cubic collision pole, producing an r-matrix with genuine spectral dependence.

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1.2. **Seven faces of $r(z)$.** The collision residue is a single object viewed through seven algebraic lenses.

Theorem 1.1 (Seven-face identification)**.** *For a modular Koszul chiral algebra $\mathcal{A}$, the following seven expressions are equal as elements of* $\mathrm{End}_{\mathcal{A}}(2)[\![z^{-1}]\!]$*:*

(F1) *The bar collision residue:* $r_{\mathcal{A}}(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$.
(F2) *The spectral R-matrix of the ordered bar:* $R(z) = \mathrm{id} + r(z)/z + \cdots$ *solves the classical Yang–Baxter equation.*
(F3) *The λ-bracket of the Poisson vertex algebra:* $r(z) = \int_0^\infty e^{-\lambda z}\{a_\lambda b\}\, d\lambda$ *(Laplace transform).*
(F4) *The Knizhnik–Zamolodchikov connection:* $\nabla_{\mathrm{KZ}} = d - r(z)\, d\log(z_1 - z_2)$.
(F5) *The Drinfeld r-matrix (for $\mathcal{A} = \widehat{\mathfrak{g}}_k$):* $r(z) = \Omega/((k+h^\vee)z)$. *Equivalently, in the trace-form convention* $r(z) = k\,\Omega_{\mathrm{tr}}/z$ *with* $k\,\Omega_{\mathrm{tr}} = \Omega/(k+h^\vee)$ *at generic k.*
(F6) *The Sklyanin bracket:* $\{L_1(u), L_2(v)\} = [r(u-v), L_1(u) \otimes L_2(v)]$.
(F7) *The Gaudin Hamiltonians:* $H_i = \sum_{j\neq i} r(z_i - z_j)$ *are mutually commuting.*

Faces (F1)–(F4) *hold for every modular Koszul algebra. Faces* (F5)–(F7) *specialize to affine Kac–Moody; the general case is obtained by replacing* $\Omega/((k+h^\vee)z)$ *with* $r_{\mathcal{A}}(z)$.

The seven faces are not seven theorems but seven readings of one equation. The collision residue $r(z) = \pi_2(\Theta_{\mathcal{A}})$ is the degree-2 projection of the universal MC element; the seven faces are the seven classical frameworks in which this projection has been studied. Drinfeld's r-matrix [**?**], the KZ connection [**?**], the Sklyanin bracket [**?**], and the Gaudin model [**?**] all extract the same binary invariant.

The relationship to the modular characteristic is immediate:

$$\kappa(\mathcal{A}) \;=\; \mathrm{av}\big(r(z)\big) \quad \text{for abelian and Virasoro-type families,} \tag{1.2}$$

The averaging map av is the Σ_2-coinvariant projection. For non-abelian affine Kac–Moody in the trace-form convention,

$$\mathrm{av}(k\,\Omega_{\mathrm{tr}}/z) \;=\; \frac{k\,\dim(\mathfrak{g})}{2h^\vee} \;=\; \kappa_{\mathrm{dp}}(\widehat{\mathfrak{g}}_k), \qquad \kappa(\widehat{\mathfrak{g}}_k) \;=\; \kappa_{\mathrm{dp}}(\widehat{\mathfrak{g}}_k) + \frac{\dim(\mathfrak{g})}{2}.$$

For Virasoro, $\mathrm{av}((c/2)/z^3 + 2T/z) = c/2$.

1.3. **DS reduction as a functor on collision residues.** Drinfeld–Sokolov reduction converts affine Kac–Moody algebras into $\mathcal{W}$-algebras: $\widehat{\mathfrak{g}}_k \xmapsto{\mathrm{DS}} \mathcal{W}^k_N(\mathfrak{g})$. At the level of the collision residue, this passage raises the pole order:

	OPE max. pole	**Collision max. pole**	**Class**
$\widehat{\mathfrak{sl}}_N$	2	1	**L**
$\mathcal{W}_N$	$2N$	$2N-1$	**M**

The Sugawara construction generates a Virasoro field $T(z)$ from the affine currents $J^a(z)$. The OPE $T(z)\,T(w)$ has a quartic pole; DS reduction extends this mechanism to all generators, producing a $\mathcal{W}$-algebra whose maximal OPE pole grows with N. The shadow depth jumps from finite (class $\mathbf{L}$) to infinite (class $\mathbf{M}$): DS reduction is the functor that transports gauge-type algebras into gravitational-type algebras.

The intertwining theorem controls the Koszul-dual behavior: the bar-cobar adjunction commutes with principal DS reduction. For non-principal reductions (hook-type in type A, proved; arbitrary nilpotent, programme), the intertwining holds on the hook-type corridor and is conjectural in full generality.

2. The Swiss-cheese realization

The bar complex of a chiral algebra on a curve X has two structures: a differential (from OPE residues on Fulton–MacPherson compactifications of X) and a coproduct (from interval-cutting in a topological direction). The bar complex $B(\mathcal{A})$ is an E_1 chiral coassociative coalgebra over $(\mathrm{ChirAss})^!$: the differential encodes holomorphic factorization on $\mathbb{C}$, the deconcatenation coproduct encodes topological factorization on $\mathbb{R}$. The two-coloured Swiss-cheese operad $\mathrm{SC}^{\mathrm{ch,top}}$ emerges on the chiral derived center pair $(C^\bullet_{\mathrm{ch}}(\mathcal{A},\mathcal{A}),\,\mathcal{A})$, not on $B(\mathcal{A})$ itself.

2.1. **The two colours.** The bar complex lives on $\overline{\mathrm{FM}}_k(\mathbf{C}) \times \mathrm{Conf}_k(\mathbb{R})$, the product of holomorphic and topological configuration spaces. This product is the operadic fingerprint of the Swiss-cheese operad:

	Closed colour	**Open colour**
Geometry	$\overline{\mathrm{FM}}_k(\mathbf{C})$	$\mathrm{Conf}_k(\mathbb{R})$
SC datum	$C^\bullet_{\mathrm{ch}}(\mathcal{A},\mathcal{A})$	$\mathcal{A}$
Bar engine	Collision-residue differential on $B(\mathcal{A})$	Deconcatenation coproduct on $B(\mathcal{A})$
Physics	Bulk operators	Boundary operators

The bar differential and the deconcatenation coproduct are the two operations of the ordered E_1 dg coalgebra $B(\mathcal{A})$. They record the holomorphic and topological directions, but they do not by themselves furnish the two colours of a Swiss-cheese algebra. The closed and open colours live on the derived-center pair $(C^\bullet_{\mathrm{ch}}(\mathcal{A},\mathcal{A}),\,\mathcal{A})$: bulk operators act on the boundary, but not conversely.

The directionality of the Swiss-cheese operad is strict: *no open inputs produce closed outputs*. Bulk operators restrict to boundary operators, but not conversely. The open colour is the E_1-direction; the passage to Σ_n-coinvariants (the closed/symmetric bar) is a lossy projection. The five main theorems A–H are the invariants that survive this projection.

2.2. **Three bar complexes.** Three bar-type objects coexist, corresponding to three cooperadic structures on the same underlying graded vector space:

(a) $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$: the *ordered bar*. Deconcatenation coproduct ($n+1$ terms), coassociative but not cocommutative. Carries the R-matrix, the KZ associator, and the full Yangian deformation. This is the E_1-coalgebra.

(b) $\overline{B}^{\Sigma}(\mathcal{A}) = \mathrm{Sym}^c(s^{-1}\bar{\mathcal{A}})$: the *symmetric bar*. Coshuffle coproduct (2^n terms), cocommutative and coassociative. This is the factorization coalgebra on $\mathrm{Ran}(X)$: the E_∞-coalgebra of Beilinson–Drinfeld.

(c) $B^{\mathrm{FG}}(\mathcal{A}) = \mathrm{Lie}^c(s^{-1}\bar{\mathcal{A}})$: the *Francis–Gaitsgory bar*. CoLie cobracket, capturing only the zeroth-pole OPE data ($a_{(0)}b$, the chiral Lie bracket). A quasi-isomorphism $B^{\mathrm{FG}} \hookrightarrow \overline{B}^{\Sigma}$ holds in characteristic zero, but B^{FG} is invisible to the curvature $\kappa(\mathcal{A})$: the modular characteristic comes from the *first* OPE pole, which the Francis–Gaitsgory bar does not see.

The natural descent is $\overline{B}^{\mathrm{ord}} \xrightarrow{\mathrm{av}} \overline{B}^{\Sigma} \hookleftarrow B^{\mathrm{FG}}$: the ordered bar maps to the symmetric bar by Σ_n-coinvariance (a lossy projection), and the Francis–Gaitsgory bar embeds into the symmetric bar (a quasi-isomorphism in characteristic zero, losing all higher-pole data). The ordered bar is the natural primitive; the other two are its images.

2.3. **PVA descent and the recognition theorem.** The cohomology of a Swiss-cheese algebra carries a Poisson vertex algebra structure: the commutative product descends from the regular OPE and the Poisson bracket from the singular OPE. This is the *PVA descent*: the classical shadow of the full quantum Swiss-cheese structure.

The six PVA axioms (Leibniz, skew-symmetry, Jacobi, sesquilinearity, ∂-bracket, and Wick) are proved geometrically from configuration-space geometry: the exchange cylinder on $\overline{\mathrm{FM}}_2(\mathbf{C})$ for skew-symmetry, the three-face Stokes theorem on $\overline{\mathrm{FM}}_3(\mathbf{C})$ for Jacobi, and boundary factorization on higher $\overline{\mathrm{FM}}_k(\mathbf{C})$ for the remaining axioms.

The recognition theorem establishes the converse:

Theorem 2.1 (Recognition)**.** *A factorization algebra on* $\mathrm{Ran}(X)$ *equipped with Weiss cosheaf descent data and compatible with the factorization isomorphisms is a* $C_*(\mathcal{W}(\mathrm{SC}^{\mathrm{ch,top}}))$*-algebra.*

The theorem converts the abstract factorization structure into a concrete operadic algebra over the Boardman–Vogt resolution of the Swiss-cheese operad. It is proved by Weiss cosheaf descent: the factorization pattern on $\mathrm{Ran}(X)$ determines the operadic composition uniquely.

The homotopy-Koszulity of $\mathrm{SC}^{\mathrm{ch,top}}$ is proved via Kontsevich formality and transfer from the classical Swiss-cheese operad (Livernet). This removes all previously conditional hypotheses: bar-cobar Quillen equivalence, filtered

Koszul duality, and the identification of line operators with dual modules all become unconditional.

3. Three-dimensional HT QFT and gauge-gravity

A four-dimensional holomorphic-topological field theory on $\mathbf{C}_z \times \mathbb{R}_t \times \mathbb{R}_{>0}$ restricts to a chiral algebra on the holomorphic boundary. The bar complex of that boundary algebra classifies twisting morphisms (couplings to the Koszul dual); the chiral derived centre computes the universal bulk; bar-cobar inversion recovers the boundary algebra itself. The modular convolution algebra organizes the full boundary-to-bulk correspondence at all genera.

3.1. **The holographic datum.** The complete data of a holographic system is a sextuple:

Definition 3.1 (Holographic modular Koszul datum)**.** The *holographic modular Koszul datum* of a chiral algebra $\mathcal{A}$ is the sextuple

$$\mathcal{H}(\mathcal{A}) \;=\; \big(\mathcal{A},\ \mathcal{A}^!,\ \mathcal{C},\ r(z),\ \Theta_{\mathcal{A}},\ \nabla^{\mathrm{hol}}\big)\,, \tag{3.1}$$

where $\mathcal{A}^! = (H^*(B(\mathcal{A})))^\vee$ is the Koszul dual, $\mathcal{C} \simeq \mathcal{A}^!\text{-}\mathsf{mod}$ is the line-operator category, $r(z)$ is the collision residue, $\Theta_{\mathcal{A}}$ is the universal MC element, and $\nabla^{\mathrm{hol}}_{g,n} = d - \mathrm{Sh}_{g,n}(\Theta_{\mathcal{A}})$ is the modular shadow connection. Every component is a projection of $\Theta_{\mathcal{A}}$.

The datum $\mathcal{H}(\mathcal{A})$ is algorithmically reconstructed from the boundary OPE of $\mathcal{A}$: $\mathcal{A} \overset{B}{\longmapsto} \Theta_{\mathcal{A}} \xmapsto{\mathrm{Res}^{\mathrm{coll}}} r(z) \xmapsto{\mathrm{RTT}} \mathcal{C}$. Each step uses only the output of the preceding step; the entire holographic system is determined by the chiral algebra on the boundary.

3.2. **The 4d–3d–2d resolution.** The three volumes of the programme correspond to three levels of a dimensional hierarchy:

d	**Level**	**Operadic type**	**Key datum**	**Volume**
4	CY source	E_2	CY category $\mathcal{C}$	III
3	Swiss-cheese	E_1	R-matrix $r(z)$	II
2	Modular shadow	E_∞	$\kappa(\mathcal{A})$	I

At each level, information is lost. From 4d to 3d, one topological direction is forgotten: $E_2 \to E_1$ (the braiding becomes ordering). From 3d to 2d, Σ_n-coinvariance projects the R-matrix to a scalar: $r(z) \mapsto \kappa$. The five main theorems are the invariants that survive both projections.

3.3. **Gauge versus gravity.** The shadow depth classification (Definition **??**) separates the standard landscape into two physical regimes:

(a) *Gauge theories* (classes **G**, **L**): finite shadow depth, collision residue with at most a simple pole. The bar complex is effectively quadratic

(class **G**) or cubic (class **L**). The representation theory is governed by quantum groups with rational R-matrices.

(b) *Gravitational theories* (classes **C**, **M**): the shadow tower extends to degree 4 (class **C**) or is infinite (class **M**). The collision residue has higher-order poles. The bar complex is genuinely non-formal: the full A_∞-tower is irreducible. The Koszul duality $\mathrm{Vir}_c^! = \mathrm{Vir}_{26-c}$ (or $\mathcal{W}_N^! = \mathcal{W}_N^{k'}$ at the dual level) exchanges matter and ghost sectors.

DS reduction is the functor connecting these regimes: it transports affine Kac–Moody (class **L**, gauge) to $\mathcal{W}$-algebras (class **M**, gravity) by raising the pole order through the Sugawara construction. The physical content: the Sugawara stress tensor converts gauge currents into a metric, and the bar complex detects this conversion as the appearance of higher collision poles.

3.4. **The critical string at $c = 26$.** The Virasoro Koszul duality $\mathrm{Vir}_c^! = \mathrm{Vir}_{26-c}$ identifies three distinguished points:

c	**Property**	κ	**Clifford**
0	Gauge point: $\kappa = 0$, flat bar	0	degenerate
13	Self-dual point: $\mathrm{Vir}_{13}^! = \mathrm{Vir}_{13}$	13/2	non-degenerate
26	Critical string: $\kappa^! = 0$	13	exterior algebra

At $c = 0$: $\kappa(\mathrm{Vir}_0) = 0$, the bar complex is flat at all genera, and the genus tower is trivial. At $c = 13$: $\kappa(\mathrm{Vir}_{13}) = 13/2 = \kappa(\mathrm{Vir}_{13}^!)$, the complementarity pairing is symmetric, and the entire shadow tower is self-dual. At $c = 26$: $\kappa(\mathrm{Vir}_0^!) = 0$, the Koszul dual is flat; the Clifford algebra of the complementarity pairing degenerates to an exterior algebra, and the matter-ghost cancellation condition $\kappa_{\mathrm{eff}} = \kappa(\mathrm{matter}) + \kappa(\mathrm{ghost}) = 0$ of the bosonic string is satisfied. The 26-dimensional critical string is the point where the Koszul dual's bar complex becomes flat: the gravitational content migrates from curvature to bar cohomology.

4. The Calabi–Yau bridge

Volumes I and II take a chiral algebra as given. The source of chiral algebras is geometry: a Calabi–Yau category $\mathcal{C}$ of dimension d produces a chiral algebra $A_{\mathcal{C}}$ on a curve X, with the CY trace realized as the modular characteristic. This is the content of Volume III.

4.1. **The functor Φ.** The construction proceeds in three steps. Given a CY_d category $\mathcal{C}$:

(1) The cyclic A_∞-structure on $\mathcal{C}$ determines a *Lie conformal algebra* $L_{\mathcal{C}}$: the cyclic bar differential gives the λ-bracket, the cyclic trace gives the invariant bilinear form, and the higher A_∞-operations give higher Lie conformal products.

(2) The *factorization envelope* $U_X^{\mathrm{fact}}(L_{\mathcal{C}})$ produces a factorization algebra on $\mathrm{Ran}(X)$, hence a chiral algebra $A_{\mathcal{C}}$ on X. The PBW theorem for factorization envelopes guarantees the expected filtration.
(3) The $\mathbb{S}^d$-framing on categorical Hochschild homology determines the operadic enhancement. For $d = 2$: the $\mathbb{S}^2$-action gives an E_2-structure with braided monoidal representation category. For $d = 3$: holomorphic Chern–Simons theory on $\mathcal{C}$ breaks E_2 to E_1, and the braided structure is recovered via the Drinfeld centre of the E_1-monoidal category.

Theorem 4.1 (CY-A$_2$)**.** *The functor Φ is constructed for $d = 2$: all three steps are proved. The modular characteristic of $A_{\mathcal{C}}$ equals the Euler characteristic: $\kappa(A_{\mathcal{C}}) = \chi^{\mathrm{CY}}(\mathcal{C})$.*

For $d = 3$, steps (1) and (2) are unconditional; step (3) is a programme conditional on the chain-level $\mathbb{S}^3$-framing. The passage from $d = 2$ to $d = 3$ replaces a geometric braiding (from $\pi_1(\mathrm{Conf}_2(\mathbb{R}^2)) = \mathbb{Z}$) with an algebraic one (from the Drinfeld centre of the E_1-monoidal representation category): this is the CY analogue of the Tannakian reconstruction of a quantum group from its module category.

4.2. **The E_1 stabilization.** For $d \geq 3$, the CY chiral algebra is E_1, not E_d. The reason is structural: the Schouten–Nijenhuis bracket on polyvector fields of $\mathbf{C}^d$ vanishes identically for $d \geq 3$, so the classical Lie conformal algebra is abelian and all noncommutative structure arises from quantization. The CoHA construction produces an ordered (associative) product from the extension correspondence $0 \to V' \to V \to V'' \to 0$, which has a preferred direction: sub before quotient. This is E_1, not E_2.

The framing obstruction lies in $\pi_d(BU)$, which is $\mathbb{Z}$ for even d and 0 for $d \equiv 1, 3, 7 \pmod 8$. The pattern repeats with period 8 by Bott periodicity. For $d = 2$: the obstruction is $c_1 \in \pi_2(BU) = \mathbb{Z}$, the braiding parameter. For $d = 3$: the obstruction vanishes ($\pi_3(BU) = 0$), so the $\mathbb{S}^3$-framing trivializes and the E_1-structure receives no additional enhancement. For $d = 4$: $\pi_4(BU) = \mathbb{Z}$ produces a shifted symplectic level (p_1), the CY$_4$ analogue of the Kac–Moody level.

4.3. **The $K3 \times E$ prototype.** The product of a K3 surface S and an elliptic curve E is the canonical CY$_3$ testing ground: simple enough to compute, rich enough to obstruct.

The lattice $\Lambda^{3,2}$ has signature $(3, 2)$; the Jacobi form $\phi_{0,1}(\tau, z)$ is the K3 elliptic genus; the Igusa cusp form Δ_5 is the Borcherds denominator identity. The modular characteristic of the chiral de Rham complex is $\kappa_{\mathrm{ch}}(K3 \times E) = 3 = \dim_{\mathbf{C}}$, verified by six independent paths. The BKM automorphic weight is a distinct quantity: $\kappa_{\mathrm{BKM}} = 5$ (the weight of Δ_5).

These are different numbers measuring different things. A single CY$_3$ admits multiple chiral algebraizations, each producing a different modular

characteristic. The κ-spectrum

$$\mathrm{Spec}_\kappa(K3 \times E) \;=\; \{2, 3, 5, 24\} \tag{4.1}$$

records four values: $\kappa_{\mathrm{cat}} = 2 = \chi(\mathcal{O}_{K3})$ (the holomorphic Euler characteristic), $\kappa_{\mathrm{ch}} = 3$ (the complex dimension), $\kappa_{\mathrm{BKM}} = 5$ (the Borcherds weight), and $\kappa_{\mathrm{fiber}} = 24$ (the lattice rank). Four algebras see four aspects of the same manifold.

Four structural obstructions separate the shadow obstruction tower from the full automorphic form:

(O1) *Categorical*: the shadow tower produces Taylor coefficients; the denominator identity is an automorphic form on $\mathcal{H}_2$.
(O2) *κ-mismatch*: $\kappa_{\mathrm{ch}} = 3 \neq 5 = \kappa_{\mathrm{BKM}}$.
(O3) *Second quantization*: the physical DT partition function is the DMVV symmetric product of single-copy data; the bar complex is the single copy.
(O4) *Schottky*: at genus $g \geq 4$, the Torelli map $\mathcal{M}_g \to \mathcal{A}_g$ is no longer dominant, and the shadow tower is imprisoned in tautological classes.

Each obstruction points to a research programme. The κ-spectrum is the first invariant that separates CY chiral algebras with the same underlying manifold.

4.4. E_1 **at** $d = 3$ **via holomorphic Chern–Simons.** For a CY threefold X, holomorphic Chern–Simons theory on X with gauge group G produces a chiral algebra A_X of E_1-type. The spectral parameter of the R-matrix is identified with the holomorphic coordinate of the 4d theory restricted to a 2d boundary. The braided monoidal structure on the representation category is recovered through the Drinfeld centre:

$$\mathcal{Z}\big(\mathrm{Rep}^{E_1}(A_X)\big) \;\simeq\; \mathrm{Rep}^{E_2}(A_X)\,, \tag{4.2}$$

where the right-hand side is the braided monoidal category of E_2-modules. For toric CY_3, the E_1-algebra is the critical CoHA $\mathrm{CoHA}(Q_X)$: the positive half of an affine super Yangian $Y(\widehat{\mathfrak{g}}_{Q_X})$. For $\mathbf{C}^3$: the Jordan quiver gives $Y(\widehat{\mathfrak{gl}}_1) \simeq \mathcal{W}_{1+\infty}$.

The dimensional reduction principle underlies the full trilogy: $4d \to 3d$ forgets the braiding ($E_2 \to E_1$); $3d \to 2d$ forgets the ordering ($E_1 \to E_\infty$ by Σ_n-coinvariance). The five main theorems live at the $2d$ level; the seven faces of the collision residue live at the $3d$ level; the CY source lives at the $4d$ level. The modular Koszul engine converts enumerative geometry (Donaldson–Thomas invariants, root multiplicities, denominator identities) into homotopy algebra (bar complexes, MC elements, genus expansions). One Maurer–Cartan element $\Theta_{A_\mathcal{C}}$ absorbs the entire automorphic datum.

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